

Appendix H. Core Health Valuation Functions in U.S. Setup

This appendix presents the core unit values that are available in BenMAP for each of the health endpoints included in the current suite of health impact functions. Specifically, this appendix includes the values currently used by U.S. EPA in regulatory impact analyses. For the .apvx files summarizing current EPA practices, see:

<https://www.epa.gov/benmap/benmap-community-edition>

Wherever possible, we present a distribution of the unit value, characterizing the uncertainty surrounding any point estimate. The mean of the distribution is taken as the point estimate of the unit value, and the distribution itself is used to characterize the uncertainty surrounding the unit value, which feeds into the uncertainty surrounding the monetary benefits associated with reducing the incidence of the health endpoint. Below we give detailed descriptions of the derivations of unit values and their distributions, as well as tables listing the unit values and their distributions, available for each health endpoint. The definitions of the distributions and their parameters are given in Table H-1.

Table H-1. Unit Value Uncertainty Distributions and Their Parameters

Distribution*	Parameter 1 (P1)	Parameter 2 (P2)
Normal	Standard deviation	–
Triangular	Minimum value	Maximum value
Lognormal **	Mean of corresponding normal distribution	Standard deviation of corresponding normal distribution
Uniform	Minimum value	Maximum value
Weibull ***	α	β

*In all cases, BenMAP calculates the mean of the distribution, which is used as the “point estimate” of the unit value.

** If Y is a normal random variable, and $Y = \log_e X$, then X is lognormally distributed. Equivalently, X is lognormally distributed if $X = e^Y$, where Y is normally distributed.

*** The Weibull distribution has the following probability density function:

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{x}{\alpha}\right)^{\beta-1} e^{-(x/\alpha)^\beta}$$

This appendix also presents EPA methods for developing income growth adjustment factors that allow BenMAP-CE users to adjust the WTP estimates to account for the growth in income over time.

H.1 Mortality

The economics literature concerning the appropriate method for valuing reductions in premature mortality risk is still developing. The adoption of a value for the projected reduction in the risk of premature mortality is the subject of continuing discussion within the economics and public policy analysis communities. Issues such as the

appropriate discount rate and whether there are factors, such as age or the quality of life, that should be taken into consideration when estimating the value of avoided premature mortality are still under discussion. BenMAP currently offers a variety of options reflecting the uncertainty surrounding the unit value for premature mortality.

H.1.1 Value of a Statistical Life Based on 26 Studies

The current undiscounted VSL used by EPA is \$8.7 million (2015\$). This estimate is the mean of a distribution fitted to 26 “value of statistical life” (VSL) estimates that appear in the economics literature and that have been identified in the Section 812 Reports to Congress as “applicable to policy analysis.” This represents an intermediate value from a variety of estimates, and it is a value EPA has frequently used in Regulatory Impact Analyses (RIAs) as well as in the Section 812 Retrospective and Prospective Analyses of the Clean Air Act. Accounting for the cessation lag, or the delay between pollutant exposure and death, this VSL equates to \$8.1 million (2015\$) using a 2% discount rate, \$7.9 million (2015\$) using a 3% discount rate and \$7.1 million (2015\$) using a 7% discount rate (U.S. EPA, 2014).

The VSL approach mirrors that of Viscusi (1992), and uses the same criteria as Viscusi in his review of value-of-life studies. The \$8.7 million estimate is consistent with Viscusi’s conclusion (updated to 2015\$) that “most of the reasonable estimates of the value of life are clustered in the \$5.2 to \$12.3 million range.” Five of the 26 studies are contingent valuation (CV) studies, which directly solicit WTP information from subjects; the rest are wage-risk studies, which base WTP estimates on estimates of the additional compensation demanded in the labor market for riskier jobs. Because this VSL-based unit value does not distinguish among people based on the age at their death or the quality of their lives, it can be applied to all premature deaths. Table H-2 presents the unit values for the 26 value-of-life studies, the 2%, 3% and 7% discounted unit values represent the core EPA values for this endpoint while the undiscounted rate represents an additional valuation function.

Table H-2. Core Unit Values for VSL based on 26-value-of-life studies

Basis for Estimate *	Age Range at Death		Unit Value (VSL) (2015\$)	Distribution of Unit Value	Parameters of Distribution	
	Min	Max			P1	P2
VSL, based on 26 value-of-life studies	0	99	8,705,114	Weibull	9,648,168	1.509588
VSL, based on 26 value-of-life studies, 2% discount rate	0	99	8,132,666	Weibull	9,013,705	1.410317
VSL, based on 26 value-of-life studies, 3% discount rate	0	99	7,887,356	Weibull	8,741,819	1.367777
VSL, based on 26 value-of-life studies, 7% discount rate	0	99	7,103,778	Weibull	7,873,354	1.231894

* The original value of a statistical life was calculated in 1990\$. We have used a factor of 1.8134, based on the All-Items CPI-U.

H.2 Hospital Admissions & Emergency Room Visits

This section presents the core values for avoided hospital admissions, as well as avoided emergency room visits. We assume that hospital admissions due to acute exposure to air pollution pass through the emergency room. However, the value of hospital admissions that we have calculated here does not account for the cost incurred in the emergency room visit.

H.2.1 Hospital Admissions

As suggested above, the total value to society of an individual's avoidance of a hospital admission can be thought of as having two components: (1) the cost of illness (COI) to society, including the total medical costs plus the value of the lost productivity, as well as (2) the WTP of the individual, as well as that of others, to avoid the pain and suffering resulting from the illness.

In the absence of estimates of social WTP to avoid hospital admissions for specific illnesses (components 1 plus 2 above), estimates of total COI (component 1) are available for use in BenMAP as conservative (lower bound) estimates. Because these estimates do not include the value of avoiding the pain and suffering resulting from the illness (component 2), they are biased downward. Some analyses adjust COI estimates upward by multiplying by an estimate of the ratio of WTP to COI, to better approximate total WTP. Other analyses have avoided making this adjustment because of the possibility of over-adjusting -- that is, possibly replacing a known downward bias with an upward bias. Based on Science Advisory Board (SAB) advice, the COI values currently available for use in BenMAP are not adjusted.

Unit values are based on ICD-code-specific estimated hospital charges and opportunity cost of time spent in the hospital (based on the average length of a hospital stay for the illness). The opportunity cost of a day spent in the hospital is estimated as the value of the lost daily wage, regardless of whether or not the individual is in the workforce.

For all hospital admissions endpoints available in BenMAP, estimates of hospital charges and lengths of hospital stays were based on discharge statistics provided by the Agency for Healthcare Research and Quality's Healthcare Utilization Project National Inpatient Sample (NIS) database (2016). The NIS is the largest inpatient care database in the United States, and it is the only national hospital database containing charge information on all patients. It contains data from a very large nationally representative sample of about eight million hospital discharges, and therefore provides the best estimates of mean hospital charges and mean lengths of stay available, with negligible standard errors. The sampling frame for the 2016 NIS is a sample of hospitals that comprises approximately 90 percent of all hospital discharges in the United States. Since the NIS is based on discharge samples, the discharge-level weight was used to weight discharges in order to produce national estimates. The principal diagnoses were used to define the health endpoints.

Since most pollution-related hospital admissions are likely unscheduled, the unit values of avoided hospital admissions used in BenMAP are based solely on unscheduled hospitalizations. The total COI for an ICD-code-specific hospital stay lasting n days is estimated as the mean hospital charge plus n times the daily lost wage.

County-specific median annual income divided by (52×5) was used to estimate county-specific median daily wage. The data source for median annual income is the 2021 American Community Survey (ACS). ACS provided data for median annual income for all individuals over 16 years old in all U.S. counties. We apply county-specific employment rates from the 2021 ACS to the mean wage data to estimate the average economic cost of a lost day or year for non-workers and for workers. For non-workers, the daily lost wage is the ACS mean daily wage net of a seven percent income tax rate, which is the percentage difference in median post-tax income and median income from U.S. Census Bureau (2021). For workers, the daily lost wage is the ACS mean daily wage scaled by a multiplier that accounts for mean county-level fringe benefits and overhead. We calculate this value using county-specific fringe benefits estimates from the U.S. Bureau of Labor Statistics' 2023 National Compensation Survey and a national overhead coefficient of 1.2 from U.S. EPA's National Center for Environmental Economics (NCEE), as described in U.S. EPA (2023). Because wage data used in BenMAP are county-specific, the unit value for a hospital admission varies from one county to another.

Although the data for hospital charges are from year 2016, the default hospital admission unit values in BenMAP are in year 2015 dollars to be consistent with the unit values of other health endpoints in BenMAP. This was done by inflating the medical costs (2016 dollars) to 2015 dollars using BenMAP's medical inflation index.

The hospital admission outcomes that the EPA uses in its regulatory analyses are given in Table H-3. Although unit values available for use in BenMAP are county-specific, the national median daily wage was used to calculate opportunity costs and total costs.

Table H-3. Core Unit Values Available for Hospital Admissions

Endpoint	ICD Codes	Age Range		Mean Hospital Charge (2015 \$)	Mean Length of Stay (days)	Total Cost of Illness (Unit Value in 2015\$)*
		Min	Max			
HA, All Cardiac Outcomes	390-459	0	99	\$16,045	5.05	\$16,918
HA, All Respiratory	460-519	0	18	\$9,075	3.49	\$9,678
HA, All Respiratory	460-519	65	99	\$35,402	6.07	\$36,451
HA, Alzheimer's Disease	331.0	65	99	\$10,696	7.95	\$12,070
HA, Cardio-, Cerebro- and Peripheral Vascular Disease	410- 414, 429, 426- 427, 428, 430-438, 440-449	65	99	\$14,665	4.82	\$15,498
HA, Respiratory-1	491, 492, 493, 496	0	99	\$7,676	3.86	\$8,343
HA, Respiratory-2	464-466, 480-487, 490-492, 493	65	99	\$9,003	4.66	\$9,808
HA, Parkinson's Disease	332	18	99	\$12,190	3.83	\$12,852

* The opportunity cost of a day spent in the hospital was estimated, for the above exhibit, at the median daily wage of all workers, regardless of age. The median daily wage was calculated by dividing the median weekly wage (\$864 in 2015\$) by 5. The median weekly wages for 2015 were obtained from the U.S. Census Bureau's 2015 American Community Survey, "Selected Economic Characteristics: 2015 American Community Survey 1-Year Estimates."

For two hospital admission endpoints, Alzheimer's and Parkinson's Disease, we calculated the lifetime cost of illness in addition to calculating the costs associated with the initial hospitalization.

Valuation sources of Alzheimer's disease lifetime medical costs were available from the Alzheimer's Association (2020) report and Jutkowitz et al. (2017). Using Alzheimer's Association (2020), we first developed an estimate of incremental annual medical expenses for Medicare beneficiaries living with Alzheimer's Disease. Then, using the estimated life expectancy duration of 5 years from Jutkowitz et al. (2017), we estimated total discounted present values for a five-year stream of costs using 2, 3 and 7 percent discount rates (Table H-4). We note that the median age of Alzheimer's disease onset is after the age of 65. As such, we exclude any potential lost wages given the low rate of labor force participation in this age group. Lifetime medical costs, excluding initial hospitalization, are estimated at \$159,934 using a 2% discount rate, \$156,920 using a 3% discount rate or \$145,946 using a 7% discount rate in 2015.

Table H-4. Alzheimer's Disease Valuation (2015\$)

Year	2% Discount Rate	3% Discount Rate	7% Discount Rate
0	\$33,266	\$33,266	\$33,266
1	\$32,614	\$32,297	\$31,090
2	\$31,974	\$31,357	\$29,056
3	\$31,347	\$30,443	\$27,155
4	\$30,733	\$29,557	\$25,379
Total Lifetime Costs	\$159,934	\$156,920	\$145,946

Estimates of lifetime costs for Parkinson's Disease were provided by Yang et al. (2020), including direct, indirect, and non-medical costs. Using Yang et al. (2020), we first developed an annual estimate of excess costs associated with living with Parkinson's Disease. Then, using the estimated life expectancy duration of 14.6 years from De Pablo-Fernandez et al., 2017, we calculated the present value of lifetime costs over this period using 2, 3 and 7 percent discount rates (Table H-5). Lifetime medical costs are estimated at \$571,886 using a 2% discount rate, \$537,409 using a 3% discount rate, and \$428,398 using a 7% discount rate in 2015\$.

Table H-5. Lifetime Parkinson's Disease Valuation Estimate Calculation

Year	2% Discount Rate	3% Discount Rate	7% Discount Rate
0	\$44,718	\$44,718	\$44,718
1	\$43,841	\$43,416	\$41,793

Year	2% Discount Rate	3% Discount Rate	7% Discount Rate
2	\$42,982	\$42,151	\$39,059
3	\$42,139	\$40,924	\$36,503
4	\$41,313	\$39,732	\$34,115
5	\$40,502	\$38,574	\$31,883
6	\$39,708	\$37,451	\$29,798
7	\$38,930	\$36,360	\$27,848
8	\$38,166	\$35,301	\$26,026
9	\$37,418	\$34,273	\$24,324
10	\$36,684	\$33,275	\$22,732
11	\$35,965	\$32,305	\$21,245
12	\$35,260	\$31,364	\$19,855
13	\$34,568	\$30,451	\$18,556
13.6	\$19,671	\$17,117	\$9,942
Total Lifetime Costs (14.6 yr survival)	\$571,886	\$537,409	\$428,398

H.2.2 Emergency Room Visits

As with hospital admissions, to value emergency room visits we develop primary COI estimates using data from the Healthcare Cost and Utilization Project (HCUP). The 2016 Nationwide Emergency Department Sample (NEDS) provides recent, nationally representative information on medical treatment in emergency departments. In the case of emergency department visits, valuation estimates include only the medical costs.

The NEDS dataset includes discharge-level observations. That is, each data point represents one individual being discharged from the emergency department (NEDS). Because individuals are treated in these settings for a variety of reasons, we use medical billing codes to extract observations related to each health endpoint. The epidemiological studies described in Appendix E, F and G provide ICD-9 codes for each illness; however, recent HCUP datasets (including NEDS) use ICD-10 codes. Thus, we first crosswalk the relevant ICD-9 codes to associated ICD-10 codes using a mapping provided by the U.S. Centers for Disease Control. We then identify all discharges in the HCUP datasets with ICD-10 codes that match to a study's ICD-9 code(s). Because HCUP datasets often include multiple ICD-10 codes for each discharge, we focus on the principal diagnosis (i.e., the first-listed ICD-10 code).

In the NIS dataset, we convert total charges (i.e., the amount billed to patients, employers, or insurance providers) into estimates of total costs (i.e., the final reimbursements for medical treatment). Unadjusted charges are not suitable for use in regulatory analysis because posted prices generally do not reflect actual medical costs due, in part, to negotiation between medical providers and payers (e.g., insurance companies). We assume that adjusted charges reflect the actual revenue the hospital

receives and thus the actual cost of providing care. This conversion is completed using hospital-specific cost-to-charge (CCR) ratios provided with NIS. Because CCRs are not available for NEDS, we apply average CCRs for each endpoint in NIS to the same set of ICD-10 codes in NEDS.

For each health endpoint, mean estimates are calculated using estimation commands for survey data to account for the sampling design and sample discharge weights of the HCUP data. This results in estimates of mean costs and a 95% confidence interval, which represents uncertainty in our valuation estimates of medical costs. The resulting estimates are presented in Table H-6.

Table H-6. Core Unit Values Available for Emergency Department Visits

Endpoint	ICD Codes	Age Range		Mean Unit Value (2015 \$)
		Min	Max	
ER visits, All Cardiac Outcomes	390-459	0	99	\$1,161
ER visits, respiratory	491-493, 460-466, 477, 480-486, 496, 786.07, 786.09	0	99	\$875

H.2.3 Emergency Room Visits for Asthma

Two unit values are currently available for use in BenMAP for asthma emergency room (ER) visits. One is \$533.69, from Smith et al., 1997, who reported that there were approximately 1.2 million asthma-related ER visits made in 1987, at a total cost of \$186.5 million, in 1987\$. The average cost per visit was therefore \$155 in 1987\$, or \$533.69 in 2015\$ (using the CPI for medical care to adjust to 2015\$). The uncertainty surrounding this estimate, based on the uncertainty surrounding the number of ER visits and the total cost of all visits reported by Smith et al. is characterized by a triangular distribution centered at \$533.69, on the interval [\$395.14, \$738.19].

A second unit value is \$446.52 from Stanford et al. (1999). This study considered asthmatics in 1996-1997, in comparison to the Smith et al. (1997) study, which used 1987 National Medical Expenditure Survey (NMES) data. In comparing their study, the authors note that the 1987 NMES, used by Smith et al., “may not reflect changes in treatment patterns during the 1990s.” In addition, its costs are the costs to the hospital (or ER) for treating asthma rather than charges or payments by the patient and/or third party payer. Costs to the ER are probably a better measure of the value of the medical resources used up on an asthma ER visit (see above for a discussion of costs versus charges).

The unit values and the corresponding distributions available in BenMAP for asthma-related ER visits are summarized in Table H-7.

Table H-7. Core Unit Values Available for Asthma-Related ER Visits

Basis for Estimate	Age Range		Unit Value (2015\$)	Distribution of Unit Value	Parameters of Distribution	
	Min	Max			P1	P2
COI: Smith et al. (1997)	0	99	\$534	Triangular	\$395	\$738
COI: Standford et al. (1999)	0	99	\$447	Normal	8.95	--

H.2.4 Emergency Hospital Admissions for All Respiratory

For HIFs based on studies that specifically assess the impact of pollutants on emergency hospital admissions (EHAs), we develop primary COI estimates based on a combination of hospitalization and emergency department visit costs using data from the Healthcare Cost and Utilization Project (HCUP). Within HCUP, we rely on both the 2016 Nationwide Emergency Department Sample (NEDS), and the 2016 Agency for Healthcare Research and Quality's Healthcare Utilization Project National Inpatient Sample (NIS) database for cost data. NEDS provides recent, nationally representative information on medical treatment in emergency departments, while NIS provides national hospital inpatient care data including charge information on all patients. Valuation estimates are based on the sum of total medical costs incurred in the emergency department and during the subsequent inpatient hospital stay plus the value of the lost productivity for time in hospital.

The NEDS and NIS datasets index cost data by medical billing codes. The principal diagnoses were used to define the health effects, see Section H.2.2 for additional details on health effect definition.

In the NIS dataset, we convert total charges (i.e., the amount billed to patients, employers, or insurance providers) into estimates of total costs (i.e., the final reimbursements for medical treatment) using hospital-specific cost-to-charge (CCR) ratios provided with NIS. Unadjusted charges are not suitable for use in regulatory analysis because posted prices generally do not reflect actual medical costs due, in part, to negotiation between medical providers and payers (e.g., insurance companies). We assume that adjusted charges reflect the actual revenue the hospital receives and thus the actual cost of providing care. Because CCRs are not available for NEDS, we apply average CCRs for each health effect in NIS to the same set of ICD-10 codes in NEDS.

For EHAs, mean cost estimates are calculated for both hospitalizations originating in the ER, via NIS data, and emergency department visits resulting in inpatient stays, via NEDS data, using estimation commands for survey data to account for the sampling design and sample discharge weights of the HCUP data. This results in estimates of mean costs and average length of hospital stay along with 95% confidence intervals, which represents uncertainty in our valuation estimates of medical costs and lost wages. The total COI for an emergency hospital stay lasting n days is estimated as the mean total ED and hospital charge plus n times the daily lost wage. Opportunity cost of a day spent in the hospital is estimated as the value of the lost daily wage, regardless of

whether or not the individual is in the workforce. Lost daily wages are described in more detail in Section H.2.1.

The resulting EHA estimates for all respiratory illnesses are presented in Table H-8. Although unit values available for use in BenMAP are county-specific, the national median daily wage was used to calculate opportunity costs and total costs.

Table H-8. Core Unit Values Available for Respiratory-related EHAs

Endpoint	ICD Codes	Age Range		Mean Hospital Charge (2015\$)	Mean Length of Stay (days)	Total Cost of Illness (2015\$)
		Min	Max			
EHA, All Respiratory	460-519	65	99	\$11,111	5.09	\$11,990

H.3 Other Health Endpoint Occurrence

Monetary valuation estimates for health endpoint occurrences other than hospital admissions or emergency department visits are described below, listed in alphabetical order.

H.3.1 Lung Cancer

The unit value for non-fatal lung cancer incidence is derived from the direct medical costs of lung cancer treatment estimated by Kaye et al. (2018). Lost earnings are assumed to be negligible because of the low labor force participation rate among the age groups at highest risk of developing lung cancer (average age of diagnosis is approximately 70 years). Lung cancer treatment costs depend to a large extent on the phase of care, with costs in the initial year of treatment (\$17,422 for males) far exceeding the continuing costs of treatment in subsequent years (\$3,269 for males). We calculated costs over a five-year span, beginning with the initial onset which occurs with a delay after exposure. The initial year's treatment cost is summed with four years of continuing annual costs discounted by 2%, 3% and 7%.

Furthermore, Kaye et al. (2018) provides separate treatment cost estimates for men and women. The distribution of new lung cancer cases by sex in the United States from Siegel et al. (2019) is approximately 51% male and 49% female. This distribution of new lung cancer cases was used to weight the sex-specific cost estimates from Kaye et al. (2018) to obtain a combined five-year cost estimate for both sexes. In order to adjust the cost estimate to 2015\$ using a medical cost index, we assume that costs presented by Kaye et al. (2018) are in 2010\$ as an approximate midpoint of the data years 2007-2012. Altogether, the cost of non-fatal lung cancer incidence over a five-year period is estimated to be \$34,155 using a 2% discount rate, \$33,809 using a 3% discount rate or \$32,548 using a 7% discount rate (Table H-9).

Table H-9. Core Unit Values Available for non-fatal Lung Cancer

Health Endpoint	Basis for Estimate	Age Range		Unit Value (2015\$)
		Min	Max	
Lung Cancer	COI: 5 yrs med, 2% DR, Kaye (2018)	30	99	\$34,155
	COI: 5 yrs med, 3% DR, Kaye (2018)	30	99	\$33,809
	COI: 5 yrs med, 7% DR, Kaye (2018)	30	99	\$32,548

For an outcome such as lung cancer, there is an expected time lag between changes in pollutant exposure in a given year and the total realization of health effect benefits, commonly referred to in regulatory analyses as the “cessation lag.” The time between exposure and diagnosis can be quite long, on the order of years to decades, to realize the full benefits of the air quality improvements. This latency period is important in order to properly discount the economic value of these health benefits.

To estimate the latency period, we performed a literature search using the keywords “non-fatal lung cancer,” “lung cancer,” “PM_{2.5},” “latency,” and “incidence.” Five papers that estimate the risk of lung cancer incidence from PM_{2.5} exposure using a latency period were identified. The latency period length and country of the identified papers are summarized in Table H-10. Based on estimates of lung cancer latency from the literature, 10 years was the most common latency period estimate found in the literature (i.e., the mode).

Table H-10. Latency Periods Used in Lung Cancer Risk Assessment Papers

Study	Latency Period (years)	Location
Gogna et al., 2019	5	Canada
Bai et al., 2020	4; 10	Canada
Kulhanova et al., 2018	10	France
Coleman et al., 2020	10; 15	US
Harrison et al., 2004	20	US

To account for the latency period between air pollution reductions and avoided lung cancer diagnoses in our economic valuation estimates, we developed an age-at-diagnosis cessation lag distribution method based on an approach previously used to estimate avoided cases of kidney cancer in analyses of water quality rules (U.S. EPA, 2017). The method uses lung and bronchus cancer diagnosis age-distribution from the Surveillance, Epidemiology, and End Results Program (SEER). For this model, we assumed that the case reduction distribution would follow the age-pattern of cancer diagnosis between the age at which the exposure change occurs and 99 years. Table H-11 shows an example case reduction distribution calculation for an exposure change experienced at 55. SEER estimates 92.2% of lung and bronchus cancer cases occur in individuals 55 years and older. Dividing the percentages in the remaining age bins by 92.2% (the percent of lung and bronchus cancer diagnoses between the age of exposure

change and end of lifetime), we find that there is a 24% chance that the risk reductions for a 55-year-old occur between ages 55 and 64, a 37% chance that the case reductions occurs between ages 65 and 74, etc. For distributing avoided cases within an age bin, we assume an equal incidence distribution across years within each bin.

Table H-11. Percent Lung and Bronchus Cancer Incidence by Age and Distribution of Risk Reduction by Age for an Exposure Change at 55

Age Group	Percent New Cases per Year by Age*	Percent of New Cases Occurring at or After Age 55 ¹
0-20	0	NA
20-34	0.2	NA
35-44	0.9	NA
45-54	6.6	NA
55-64	21.8	24
65-74	34.1	37
75-84	26.6	29
85-99	9.7	11
55-99	92.2	100

*May not sum to 100% due to rounding

¹Calculated as the percentage in column 2 divided by 92.2%, where 92.2% is the percentage of lung and bronchus incidence between age 55 and 99.

H.3.2 Out of Hospital Cardiac Arrest

The COI for cardiac arrests occurring outside of the hospital is derived from O'Sullivan et al. (2011), who estimate three-year medical costs associated with cardiovascular disease events among adults ages 35 and older in the U.S. The authors rely on administrative claims data from a large U.S. health plan and develop econometric models to predict medical costs for 15 different cardiovascular events, including cardiac arrest, referred to as resuscitated cardiac arrest. The dataset includes over 20 million commercial and Medical Advantage members between 2002 and 2006. Cardiac arrests are identified using the ICD-9 code 427.5. The authors use propensity score matching to develop a control group with which to compare costs versus individuals that suffered cardiac arrest. Medical costs occurring within the month of the event were excluded to avoid double counting hospitalization costs, which are separately captured by the hospitalization valuation functions. Over three years, the total medical costs, excluding hospitalization, are \$36,142 (undiscounted, inflated to 2015\$), or \$35,880 using a 2% discount rate, \$35,753 using a 3% discount rate and \$35,282 for a 7% discount rate (Table H-12).

Table H-12. Valuation Estimate for Cardiac Arrests (2015\$)

Costs	Cumulative Costs	Annual Costs			
		Undiscounted	2% Discount Rate	3% Discount Rate	7% Discount Rate
<i>Month of Event*</i>	\$43,904	\$43,904	\$43,904	\$43,904	\$43,904
Year 1	\$71,901	\$27,997	\$27,997	\$27,997	\$27,997
Year 2	\$74,701	\$2,800	\$2,745	\$2,718	\$2,617
Year 3	\$80,046	\$5,345	\$5,138	\$5,038	\$4,668
Years 1-3	\$80,046	\$36,142	\$35,880	\$35,753	\$35,282

*Excluded to avoid double-counting with hospitalization costs.

H.3.3 Stroke

Maniloff and Fann (2023) developed COI estimates of direct and delayed health care costs, loss of wage income, and home paid caregiving costs associated with a non-fatal first stroke. The authors conduct a regression analysis based on 2017 – 2019 medical expenditure data from the Agency for Healthcare Research and Quality’s (AHRQ’s) Medical Expenditure Panel Survey (MEPS) to estimate the difference in total medical expenditures between individuals who report having been diagnosed with new onset stroke and otherwise similar individuals who do report a stroke diagnosis. This approach allows for inclusion of direct medical costs due to the stroke itself and medical costs due subsequent health effects that may be related to having experienced a stroke (sequelae). The MEPS data includes actual (not billed) insurer payments, government payments and individual payments, and includes agency and non-agency paid caregiving⁴² in its caregiving cost estimates. The authors report the annual costs and present value of lifetime costs of a first stroke for those ages 18 and older, broken into three cost categories: health care, lost wage income, and home caregiving. The authors estimated lifetime costs using data on expected remaining life expectancy at age of first stroke; the resulting present value lifetime estimates are the values used in the BenMAP valuation function. The study estimated the average present value lifetime COI for a new onset stroke to be \$159,067 (2015\$) using a 2% discount rate, \$150,675 (2015\$) using a 3% discount rate and \$124,845 (2015\$) using a 7% discount rate.

H.4 Acute Symptoms and Illness Not Requiring Hospitalization

Several acute symptoms and illnesses have been associated with air pollution, including acute bronchitis in children, upper and lower respiratory symptoms, and exacerbation of asthma (as indicated by one of several symptoms whose occurrence in an asthmatic generally suggests the onset of an asthma episode). In addition, several more general health endpoints which are associated with one or more of these acute symptoms and illnesses, such as minor restricted activity days, school loss days, and work loss days, have also been associated with air pollution. We briefly discuss the derivation of the

⁴² Agency caregivers are individuals who are employed by an established home care agency, whereas non-agency workers include private and independent at-home caregivers.

unit values for acute respiratory symptoms (minor restricted activity days), asthma exacerbation, and school loss days. Tables H-13 and H-14 summarize the values used by EPA in their regulatory impact analyses.

Table H-13. Additional Unit Values Available for Myocardial Infarction

Basis of Estimate	Age Range		Medical Cost (2015\$)	Opportunity Cost (2015\$)	Total Cost (2015\$)
	Min	Max			
COI: present value of medical and opportunity wage costs over estimated years of life at the time of diagnosis, 2% DR, Maniloff & Fann (2023)	18	44	\$167,000	\$368,000 (\$139,000 - \$1,100,000)	\$539,000 (\$306,000 - \$1,300,000)
	45	54	\$83,000	\$218,000 (\$82,000 - \$674,000)	\$301,000 (\$166,000 - \$757,000)
	55	64	\$57,000	\$148,000 (\$56,000 - \$456,000)	\$204,000 (\$112,000 - \$512,000)
	65	99	\$45,000	\$86,000 (\$32,000 - \$265,000)	\$131,000 (\$78,000 - \$311,000)
COI: present value of medical and opportunity wage costs over estimated years of life at the time of diagnosis, 3% DR, Maniloff & Fann (2023)	18	44	\$151,000	\$334,000 (\$126,000 - \$1,033,000)	\$485,000 (\$277,000 - \$1,184,000)
	45	54	\$78,000	\$203,000 (\$77,000 - \$628,000)	\$281,000 (\$155,000 - \$706,000)
	55	64	\$53,000	\$140,000 (\$53,000 - \$432,000)	\$193,000 (\$106,000 - \$485,000)
	65	99	\$44,000	\$83,000 (\$31,000 - \$255,000)	\$127,000 (\$75,000 - \$299,000)
COI: present value of medical and opportunity wage costs over estimated years of life at the time of diagnosis, 7% DR, Maniloff & Fann (2023)	18	44	\$108,000	\$238,000 (\$90,000 - \$734,000)	\$346,000 (\$198,000 - \$842,000)
	45	54	\$61,000	\$160,000 (\$60,000 - \$495,000)	\$222,000 (\$122,000 - \$556,000)
	55	64	\$44,000	\$114,000 (\$43,000 - \$353,000)	\$158,000 (\$87,000 - \$397,000)
	65	99	\$38,000	\$72,000 (\$27,000 - \$224,000)	\$111,000 (\$66,000 - \$262,000)

* Costs are rounded to the nearest thousand dollars. Because we scale the wage loss estimates from Maniloff & Fann (2023) using county-level workers' net costs and labor force participation rates, the values for the opportunity wage costs are not constant and vary by county. Consequently, the total costs also vary by county. For the opportunity costs and total

costs, we present the average costs across all counties and a range with the minimum and maximum values (in parentheses).

Table H-14. Core Unit Values Available for Acute Symptoms and Illnesses

Health Endpoint	Basis for Estimate	Age Range		Unit Value (2015\$)	Distribution of Unit Value	Parameters of Distribution	
		Min	Max			P1	P2
Minor Restricted Activity Days	WTP: 1 day, CV studies	18	99	\$70	Triangular	28.51	110.62
New Onset Asthma	COI: lifetime med, lifetime productivity, 2% DR	18	99	\$182,681	None	0	0
	COI: lifetime med, lifetime productivity, 3% DR	18	99	\$146,370	None	0	0
	COI: lifetime med, lifetime productivity, 7% DR	18	99	\$77,490	None	0	0
Cough; Chest Tightness; Shortness of Breath; Wheeze	WTP: 1 symptom-day, Dickie and Mesmen (2004)	0	17	\$219	LogNormal	5.390	0.078
		18	99	\$115	LogNormal	5.390	0.078
Asthma Symptoms (Albuterol Use), Albuterol Use	COI: use of inhaler	0	99	\$0.35	None	0	0
Allergic Rhinitis	COI: 1 yr med costs	0	17	\$600	None	0	0
Work Loss Days *	County average of mean daily wage	18	65	\$298*	None	N/A	N/A
School Loss Days	County average of caregiver costs and loss of learning, 2% DR	5	6	\$4,056*	None	N/A	N/A
		7	12	\$1,235*	None	N/A	N/A
		13	18	\$903*	None	N/A	N/A
	County average of caregiver costs and loss of learning, 3% DR	5	6	\$3,070*	None	N/A	N/A
		7	12	\$977*	None	N/A	N/A
		13	18	\$731*	None	N/A	N/A
	County average of caregiver costs and loss of learning, 7% DR	5	6	\$1,265*	None	N/A	N/A
		7	12	\$506*	None	N/A	N/A
		13	18	\$417*	None	N/A	N/A

* The applied unit values in BenMAP for these measures are county-specific, based on county-specific mean wages, fringe benefits, and/or employment rates. The unit values shown in this table reflect the county average of the county-specific variables applied to these health outcomes.

H.4.1 Non-Fatal Myocardial Infarctions (Heart Attacks)

In the absence of a suitable WTP value for reductions in the risk of non-fatal heart attacks, BenMAP relies on a cost-of-illness (COI) valuation approach. This approach includes two components: (1) direct medical costs and (2) opportunity costs associated with lost earnings due to illness.

The updated valuation function draws on estimates from Maniloff and Fann (2023), who provide age-specific present value estimates of costs associated with non-fatal acute myocardial infarctions (AMIs). These costs include both direct medical expenditures and wage-related opportunity costs and are calculated over the expected remaining years of life at the time of diagnosis, using 2%, 3% and 7% discount rates. The authors provide results for the following age groups: 18–44, 45–54, 55–64, and 65–99. These age-specific values are applied directly in BenMAP to estimate the economic benefits of avoided AMIs.

Maniloff and Fann use household-level data from the Medical Expenditure Panel Survey (MEPS) from 2017–2019, covering over 67,000 individuals, to estimate annual impacts of AMIs on medical expenditures and wage income. Individuals who report having had a myocardial infarction are compared to otherwise similar individuals using a regression framework that controls for age, gender, cancer diagnoses, and other factors. This approach captures not only the direct medical costs of the AMI but also its broader sequelae (e.g., post-event depression or disability) that affect future earnings.

To calculate present value estimates, the authors project average annual cost differences over the estimated number of years lived after diagnosis for each age group. For myocardial infarction, the expected years of life remaining are estimated using either (1) observed years since diagnosis from survey data (conservative estimate), or (2) national life expectancy from the Centers for Disease Control and Prevention (CDC) minus the average age of diagnosis.

To account for local economic conditions, the wage loss estimates from Maniloff and Fann are scaled using county-level wages and labor force participation rates derived from the 2021 American Community Survey (ACS) 5-Year Estimates. Medical and wage-related costs are not assigned to individuals under age 18, consistent with the age coverage in the source data. Table H-13 presents the updated AMI valuation function by age group, disaggregated into total, medical, and wage components.

H.4.2 Minor Restricted Activity Days (MRADs)

Two unit values are currently available in BenMAP for MRADs associated with acute respiratory symptoms. No studies are reported to have estimated WTP to avoid a minor restricted activity day (MRAD). Although Ostro and Rothschild (1989) estimated the relationship between PM_{2.5} and MRADs, rather than minor respiratory-restricted

activity days (MRRADs, a component of MRADs), it is likely that most of the MRADs associated with exposure to PM_{2.5} are in fact MRRADs. The original unit value, then, assumes that MRADs associated with PM exposure may be more specifically defined as MRRADs, and uses the estimate of mean WTP to avoid a MRRAD.

IEc (1993) derived an estimate of WTP to avoid a MRRAD, using WTP estimates from Tolley et al. (1986) for avoiding a three-symptom combination of coughing, throat congestion, and sinusitis. This estimate of WTP to avoid a MRRAD, so defined, is \$38.37 (1990 \$).

Any estimate of mean WTP to avoid a MRRAD (or any other type of restricted activity day other than WLD) will be somewhat arbitrary because the endpoint itself is not precisely defined. Many different combinations of symptoms could presumably result in some minor or less minor restriction in activity. Krupnick and Kopp (1988) argued that mild symptoms will not be sufficient to result in a MRRAD, so that WTP to avoid a MRRAD should exceed WTP to avoid any single mild symptom. A single severe symptom or a combination of symptoms could, however, be sufficient to restrict activity. Therefore, WTP to avoid a MRRAD should, these authors argue, not necessarily exceed WTP to avoid a single severe symptom or a combination of symptoms. The “severity” of a symptom, however, is similarly not precisely defined; moreover, one level of severity of a symptom could induce restriction of activity for one individual while not doing so for another. The same is true for any particular combination of symptoms.

Given that there is inherently a substantial degree of arbitrariness in any point estimate of WTP to avoid a MRRAD (or other kinds of restricted activity days), the reasonable bounds on such an estimate must be considered. By definition, a MRRAD does not result in loss of work. WTP to avoid a MRRAD should therefore be less than WTP to avoid a WLD. At the other extreme, WTP to avoid a MRRAD should exceed WTP to avoid a single mild symptom. The highest IEC midrange estimate of WTP to avoid a single symptom is \$20.03 (1999 \$), for eye irritation. The point estimate of WTP to avoid a WLD in the benefit analysis is \$83 (1990 \$). If all the single symptoms evaluated by the studies are not severe, then the estimate of WTP to avoid a MRRAD should be somewhere between \$16 and \$83. Because the IEC estimate of \$38 falls within this range (and acknowledging the degree of arbitrariness associated with any estimate within this range), the IEC estimate is used as the mean of a triangular distribution centered at \$38, ranging from \$16 to \$61. Adjusting to 2015\$, this is a triangular distribution centered at \$69.58, ranging from \$29 to \$111.

The estimate for the MRADs that is used in EPA benefits analyses can be found in Table H-14.

H.4.3 New Onset Asthma

Maniloff and Fann (2023) developed COI estimates of direct and delayed health care costs, loss of wage income, and home paid caregiving costs associated with new onset asthma. The authors conducted a regression analysis based on 2017 – 2019 medical

expenditure data from the Agency for Healthcare Research and Quality's (AHRQ's) Medical Expenditure Panel Survey (MEPS) to estimate the difference in total medical expenditures between individuals who report having been diagnosed with new onset asthma and otherwise similar individuals who do not report an asthma diagnosis. This approach allows for inclusion of direct medical costs due to the asthma itself and medical costs due to subsequent health effects that may be related to having experienced asthma symptoms. The MEPS data includes actual (not billed) insurer payments, government payments and individual payments, and includes agency and non-agency paid caregiving in its caregiving cost estimates. The authors report the annual costs and present value of lifetime costs of new onset asthma for those ages 18 and older, broken into three cost categories: health care, lost wage income, and home caregiving. The authors estimated lifetime costs using data on expected remaining life expectancy at age of new onset asthma; the resulting present value lifetime estimates are the values used in the BenMAP valuation function. The study estimated the average present value lifetime COI for a new onset asthma to be \$182,681 (2015\$) using a 2% discount rate, \$146,370 (2015\$) using a 3% discount rate, and \$77,490 (2015\$) using a 7% discount rate.

H.4.4 Asthma Symptoms

The valuation estimates for cough, wheeze, chest tightness, and shortness of breath were estimated from the Dickie and Messman (2004) analysis of parents' WTP to relieve asthma symptoms in children and adults. The authors derive the WTP estimates from an attribute-based, stated-choice question assessing preferences to avoid acute illness as part of a survey performed in Hattiesburg, Mississippi in 2000. Survey respondents are asked to identify whether they or their child have experienced the following asthma symptoms in the past year: cough with phlegm, shortness of breath with wheezing, chest pain on deep inspiration, and fever with muscle pain and fatigue. Respondents were then assigned one of sixteen illness profiles varying by symptom, symptom duration, in days, as well as discomfort level. Dickie and Messman (2004) calculate the WTP for children ages zero to seventeen as \$219, for one avoided mild symptom-day (2015\$). The authors also provide WTP estimates by symptom, however, they represent six avoided symptom-days. Therefore, we apply the same WTP value, for one avoided mild symptom-day, to each asthma symptom endpoint.

We calculated the economic value for albuterol use associated with asthma symptoms through prescription costs for albuterol inhalers. Epocrates and GoodRx provide cost and actuation information for four common types of albuterol inhalers in 2020 dollars. Both online resources utilize published price lists, purchases, claim records, and pharmaceutical data to provide clinical statistics. Epocrates and the FDA provide cost and actuation information for one additional, less common, albuterol inhaler.⁷⁰ We divide the cost of inhalers by the actuations per inhaler to calculate an average cost per actuation across all inhaler types. We then adjust the values to 2015\$ using the Consumer Price Index (CPI) for medical care. Since medical cost index data were unavailable for 2020 at the time of these calculations, we used the most recently

available index (2019). The resulting value for asthma symptoms, albuterol use is \$0.35 per actuation (2015\$).

Table H-14 summarizes the unit values utilized by EPA for asthma related health effects.

H.4.5 Allergic Rhinitis

We derived COI estimates for allergic rhinitis (also referred to as hay fever) from the 2005 unit cost data presented by Soni (2008). The study utilizes data from the Medical Expenditure Panel Survey (MEPS) and identifies allergic rhinitis using ICD-9 code 477. Soni (2008) analyzes medical expenditures stratified by age group for the years 2000 and 2005, and calculate the cost-of-illness as the mean expenditures per person for ambulatory care, in-patient services, and prescription medications. The resulting COI for allergic rhinitis is \$600 for ages zero to seventeen (2015\$; Table H-14). The COI estimate represents mean annual medical costs for patients with hay fever. Given that the health impact function for this endpoint relates to allergic rhinitis prevalence, these estimates are more applicable than values representing only first-year costs.

H.4.6 Work Loss Days (WLDs)

Work loss days are valued at a day's wage. BenMAP calculates county-specific mean daily wages from county-specific annual wages by dividing by (52*5), then applies a multiplier of 1.46 for fringe benefits estimates based on the U.S. Bureau of Labor Statistics' 2023 National Compensation Survey and an overhead multiplier of 1.2 from U.S. EPA's National Center for Environmental Economics (NCEE) as described in U.S. EPA (2023) for workers as defined by county-specific employment rate data from the 5-year 2021 American Community Survey. The average of this county-specific daily wage measure across all counties is approximately \$298 for ages 18 to 65 (2015\$, Table H-14).

H.4.7 School Loss Days

We include two costs of school loss days: costs to the caregiver and costs related to the loss of learning for the student not attending school, where the latter are estimated as losses to future lifetime earnings. We calculate each separately and then sum them in the valuation function. We present three alternatives for this function representing three discount rates applied to lifetime earnings portion: 2 percent, 3 percent, and 7 percent.

The caregiver costs assume that an adult caregiver stays home with the child and loses any wage income they would have earned that day. For working caregivers, we follow EPA guidance and value their time at the average wage including fringe benefits and overhead costs (U.S. EPA, 2020a). We apply a multiplier of 1.46 for fringe benefits based on the U.S. Bureau of Labor Statistics' 2023 National Compensation Survey and 1.2 for overhead from U.S. EPA's National Center for Environmental Economics (NCEE) to county-specific mean wage estimates from the U.S. Census Bureau's 2021 5-year American Community Survey (ACS). For nonworking caregivers, we assume that the

opportunity cost of time is the average after tax earnings. We estimate the income tax rate for a median household to be 7% and apply to 2021 ACS county-specific mean wage estimates. The income tax rate of 7% is the percentage difference in median post-tax income and median income from (U.S. Census Bureau, 2022). The probability that a parent is working is measured with the employment population ratio among people with their own children under 18 and is 80.7% (US Bureau of Labor Statistics, 2025). We apply the employment population ratio to the proportion of the total workforce that are employed parents. The average of this county-specific measure across all counties is approximately \$239 (2015\$).

To measure the loss of learning, we update findings of Gottfried & Ansari (2022), Aucejo & Romano (2016), and Liu et al. (2021). Gottfried & Ansari (2022) estimated the impact of school absences on learning as measured by end-of-grade math test scores for grades K-1. They found a one standard deviation increase in absences for a single student reduced that student's math test score by 0.11 standard deviations. We convert this result to estimate the effect on math test scores from a single absence by dividing by the standard deviation of absences, finding that a single absence results in a decrease in math test score of 0.023 standard deviations. Aucejo & Romano (2016) estimated the same for grades 3-5 and found a decrease of 0.006 standard deviations. Similarly, Liu et al. (2021) found a decrease of 0.004 standard deviations for grades 6-11.

We multiply the estimates of the effect of school absences on learning by an estimate of the effect of learning on lifetime earnings to calculate the effect of a single absence on lifetime earnings.

For BenMAP application, we create valuation functions that use the sum of caregiver costs and loss of learning for elementary school, middle school, and high school students. We estimate county-specific averages of the combined effect by summing the county-specific estimates of caregiver costs and age-specific costs per one lost school day with 2%, 3%, and 7% discounting. The average of this county-specific measure across all counties is approximately \$3,817 (discounted at 2%), \$2,831 (discounted at 3%) and \$1,026 (discounted at 7%) per school day for ages 5 to 6 (2015\$, Table H-14). For ages 7-12, the average is approximately \$996 (discounted at 2%), \$738 (discounted at 3%) and \$268 (discounted at 7%) per school day. Finally, for ages 13-18, the average is approximately \$664 (discounted at 2%), \$492 (discounted at 3%) and \$178 (discounted at 7%) per school day.

A unit value based on the approach described above is likely to understate the value of a school loss day in at least three ways:

- 1) It omits WTP to avoid the symptoms/illness which resulted in the school absence
- 2) The approach omits other aspects of school attendance such as social and emotional development or meals
- 3) It does not account for deleterious effects on student learning in other subjects.

H.5 Developing Income Growth Adjustment Factors

Chapter 4 of the BenMAP-CE User Manual provides instructions for formatting and adding income growth data. These values are used to adjust WTP estimates for growth in real income. As discussed in that chapter, evidence and theory suggest that WTP should increase as real income increases. When reviewing the economic literature to develop income growth adjustment factors, it is important to have an economist assist. For an overview of valuation, see Chapter 7: Aggregating, Pooling, and Valuing.

Adjusting WTP to reflect growth in real income requires three steps:

1. *Identify relevant income elasticity estimates from the peer-reviewed literature.*
2. *Calculate changes in future income.*
3. *Calculate adjustments to WTP based on changes in future income and income elasticity estimates.*

1. Identifying income elasticity estimates

Income elasticity estimates relate changes in demand for goods to changes in income. Positive income elasticity suggests that as income rises, demand for the good also rises. Negative income elasticity suggests that as income rises, demand for the good falls. BenMAP-CE does not adjust Cost-of-Illness (COI) estimates according to changes in income elasticity due to the fact that COI estimates the direct cost of a health outcome; instead we adjust this metric using inflation factors described above. BenMAP-CE includes income elasticity estimates specific to the type of health endpoint associated with the WTP estimate. BenMAP-CE contains elasticity estimates for three types of health effects: minor, severe and mortality. Minor health effects are those of short duration. Severe, or chronic, health effects are of longer duration. Consistent with economic theory, the peer reviewed literature indicates that income elasticity varies according to the severity of the health effect. A review of the literature revealed a range of income elasticity estimates that varied across the studies and according to the severity of health endpoint. Table H-15 summarizes the income elasticity estimates for minor health effects, severe health effects and mortality. Here we have provided a lower-, upper- and central-elasticity estimate for each type of health endpoint.

Table H-15. Income Elasticity Estimates

Health Endpoint	Lower Bound	Central Estimate	Upper Bound
Minor Health Effect	0.04	0.15	0.30
Severe and Chronic Health Effects	0.25	0.45	0.60
Mortality	0.08	0.40	1.00

2. Calculating changes in future income

The next input to the WTP adjustment is annual changes in income. Historical US Gross Domestic Product (GDP) data (1990-2016) comes from the U.S. Bureau of Commerce's Bureau of Economic Analysis (BEA). GDP values were adjusted for inflation using the BEA's price index for GDP. We divided historical GDP values by populations provided by the BEA to estimate GDP per capita to maintain internal consistency in the calculation. Future changes in annual income are based on data presented in the Annual Energy Outlook (AEO) 2020, a report prepared by the U.S. Energy Information Administration (EIA) (AEO, 2020). AEO published annual GDP projections through the year 2050, which were adjusted for inflation using the GDP Chain-type Price Index reported by AEO. We divided projected GDP values by AEO's population projections to estimate per capita GDP, again maintaining internal consistency in the calculation.

3. Calculating changes in WTP

The income elasticity estimates from Table H-15 and the estimated changes in future income may then be used to estimate changes in future WTP for each health endpoint. The adjustment formula follows four steps:

- 1)
$$\varepsilon = \frac{\frac{\Delta WTP}{WTP}}{\frac{\Delta I}{I}} = \frac{(WTP_2 - WTP_1) \times (I_2 + I_1)}{(I_2 - I_1) \times (WTP_2 + WTP_1)}$$
- 2)
$$\begin{aligned} \varepsilon I_2 WTP_2 + \varepsilon I_2 WTP_1 - \varepsilon I_1 WTP_2 - \varepsilon I_1 WTP_1 \\ = I_2 WTP_2 + I_1 WTP_2 - I_2 WTP_1 - I_1 WTP_1 \end{aligned}$$
- 3)
$$WTP_2 \times (\varepsilon I_2 - \varepsilon I_1 - I_2 - I_1) = WTP_1 \times (\varepsilon I_1 - \varepsilon I_2 - I_1 - I_2)$$
- 4)
$$WTP_2 = WTP_1 \times \frac{\varepsilon I_1 - \varepsilon I_2 - I_1 - I_2}{\varepsilon I_2 - \varepsilon I_1 - I_2 - I_1}$$

Table H-16 summarizes the income-based WTP adjustments used within BenMAP-CE for minor health endpoints, severe health endpoints, and premature mortality. BenMAP-CE applies the "mid" income growth adjustment to the WTP for each corresponding health endpoint. The "low" and "upper" are provided for bounding the "mid" estimate.

Table H-16. Income-Based WTP Adjustments by Health Effect and Year

Year	Minor Health Endpoint			Severe Health Endpoint			Mortality		
	Low	Mid	Upper	Low	Mid	Upper	Low	Mid	Upper
1990	1	1	1	1	1	1	1	1	1
1991	0.999425	0.997845	0.995695	0.996411	0.99355	0.991409	0.99885	0.994264	0.985722
1992	1.000278	1.001043	1.002086	1.001738	1.003131	1.004177	1.000556	1.002783	1.006971
1993	1.000845	1.003171	1.006353	1.005291	1.009545	1.012746	1.00169	1.00848	1.021335
1994	1.001941	1.007299	1.014651	1.012194	1.022057	1.029519	1.003886	1.019583	1.049686
1995	1.002529	1.009516	1.019122	1.01591	1.028821	1.038614	1.005064	1.025578	1.065196
1996	1.003545	1.01336	1.026899	1.022366	1.040622	1.054532	1.007103	1.036027	1.092569
1997	1.004809	1.018155	1.036642	1.030442	1.055472	1.074652	1.009642	1.049157	1.127596
1998	1.006098	1.023062	1.046661	1.038734	1.070818	1.095552	1.012234	1.062703	1.164493
1999	1.007498	1.02841	1.057639	1.047803	1.087723	1.1187	1.015052	1.077598	1.205983
2000	1.008676	1.032928	1.066959	1.055489	1.102148	1.138556	1.017428	1.090286	1.242108
2001	1.008675	1.032924	1.06695	1.055482	1.102134	1.138537	1.017426	1.090274	1.242075
2002	1.008984	1.03411	1.069403	1.057504	1.105943	1.143797	1.018048	1.093621	1.251727
2003	1.009739	1.037019	1.075433	1.062468	1.115325	1.15678	1.019574	1.101858	1.275708
2004	1.010863	1.041352	1.084451	1.069884	1.129408	1.176348	1.021844	1.114208	1.312269
2005	1.011864	1.045228	1.092549	1.076534	1.142112	1.194079	1.02387	1.125332	1.345838
2006	1.012602	1.04809	1.09855	1.081456	1.151559	1.207315	1.025363	1.133594	1.371175
2007	1.012957	1.049471	1.101452	1.083835	1.156139	1.213747	1.026084	1.137596	1.383575
2008	1.012532	1.047821	1.097984	1.080992	1.150667	1.206063	1.025223	1.132814	1.368769
2009	1.011166	1.042525	1.086897	1.071894	1.13324	1.181689	1.022457	1.117566	1.322336
2010	1.011843	1.045146	1.092377	1.076393	1.141841	1.1937	1.023827	1.125095	1.345117
2011	1.012165	1.046395	1.094994	1.07854	1.145958	1.199463	1.024479	1.128697	1.356115
2012	1.012764	1.04872	1.099873	1.082541	1.153646	1.210245	1.025692	1.135418	1.376817
2013	1.013213	1.050467	1.103547	1.085551	1.159447	1.2184	1.026602	1.140487	1.392581
2014	1.013911	1.053184	1.109275	1.090241	1.168516	1.231181	1.028017	1.148402	1.417472
2015	1.014754	1.056473	1.116227	1.095927	1.179559	1.246797	1.029727	1.158031	1.4482
2016	1.015112	1.057873	1.119194	1.098352	1.184283	1.253496	1.030454	1.162147	1.461489
2017	1.015291	1.058571	1.120676	1.099562	1.186645	1.25685	1.030817	1.164204	1.468167
2018	1.016092	1.061712	1.127353	1.105013	1.197312	1.272029	1.032446	1.173487	1.498591
2019	1.016945	1.06506	1.134495	1.110837	1.208762	1.288383	1.03418	1.183439	1.531756
2020	1.017494	1.067221	1.139119	1.114604	1.216197	1.299038	1.035298	1.189894	1.553577
2021	1.017858	1.068652	1.142185	1.1171	1.221138	1.306133	1.036037	1.194181	1.568204
2022	1.018224	1.070097	1.145287	1.119624	1.226144	1.313335	1.036784	1.198523	1.583131
2023	1.018585	1.071523	1.148352	1.122117	1.231098	1.320474	1.03752	1.202817	1.598006
2024	1.018991	1.073127	1.151804	1.124923	1.236689	1.328546	1.038347	1.20766	1.61492
2025	1.019446	1.074925	1.155682	1.128074	1.242981	1.337648	1.039273	1.213106	1.634116
2026	1.019915	1.076785	1.159702	1.131337	1.249515	1.347122	1.040231	1.218759	1.654237
2027	1.02041	1.078749	1.163955	1.134787	1.256443	1.357191	1.041242	1.224747	1.675778
2028	1.020937	1.080842	1.168495	1.138469	1.263858	1.367993	1.042318	1.231152	1.699074
2029	1.021436	1.082827	1.17281	1.141964	1.270919	1.378307	1.043337	1.237246	1.721494
2030	1.021959	1.08491	1.17735	1.145639	1.278367	1.389214	1.044406	1.243669	1.745395
2031	1.022479	1.086985	1.181879	1.149303	1.285815	1.400148	1.045469	1.250087	1.76956
2032	1.022969	1.088943	1.186166	1.152768	1.292879	1.410546	1.046472	1.256169	1.792727
2033	1.023487	1.091016	1.190711	1.15644	1.300389	1.421627	1.047532	1.262629	1.817622
2034	1.024004	1.093088	1.195265	1.160115	1.307929	1.432783	1.048592	1.269111	1.842903
2035	1.024493	1.095052	1.199589	1.163603	1.315106	1.443429	1.049594	1.275275	1.867234
2036	1.02498	1.097006	1.203902	1.16708	1.322282	1.454101	1.050592	1.281434	1.891828
2037	1.025478	1.099009	1.208332	1.170647	1.329668	1.465113	1.051613	1.287768	1.917424
2038	1.025984	1.101051	1.212857	1.174289	1.33723	1.476418	1.052652	1.294248	1.943931
2039	1.026492	1.103099	1.217405	1.177947	1.34485	1.487839	1.053694	1.300771	1.970954
2040	1.027004	1.105171	1.222018	1.181654	1.352597	1.499482	1.054748	1.307398	1.998757
2041	1.027508	1.107212	1.226569	1.185309	1.36026	1.511031	1.055784	1.313948	2.026591
2042	1.02803	1.109326	1.231295	1.189101	1.368236	1.523084	1.056857	1.320759	2.055917
2043	1.028555	1.111457	1.236071	1.19293	1.376316	1.535329	1.057937	1.327653	2.086006
2044	1.029075	1.113573	1.240821	1.196736	1.384373	1.547576	1.059008	1.334521	2.116397
2045	1.029605	1.115728	1.245672	1.20062	1.392623	1.560152	1.060099	1.341548	2.147923
2046	1.030116	1.117812	1.250373	1.20438	1.400637	1.572403	1.061152	1.348368	2.178951
2047	1.030638	1.119946	1.255197	1.208236	1.408882	1.585045	1.06223	1.355378	2.211295
2048	1.031135	1.121978	1.259801	1.211913	1.416771	1.597176	1.063255	1.36208	2.242655
2049	1.03162	1.123965	1.264312	1.215513	1.424519	1.609122	1.064256	1.368656	2.273847
2050	1.032088	1.125883	1.268677	1.218994	1.432035	1.620744	1.065222	1.37503	2.304491

H.6 Inflation Indices

Chapter 4 of the BenMAP-CE User Manual provides instructions for formatting and adding inflation data. These values are used to adjust economic values to express monetary units in a consistent dollar year. As discussed in that chapter, BenMAP-CE includes inflation factors for three different types of values. The source for these values is included in Table H-17. These values were re-indexed to \$2015 prior to import in BenMAP-CE.

Table H-17. Inflation Factors

Name	Description	Years	Source
All Goods Index	Value of generic goods and services	1980-2022	BLS, Data Series CUUR0000SA0 at http://data.bls.gov/cgi-bin/surveymost?cu
Medical Cost Index	Value of medical expenses		BLS, Data Series CUUR0000SAM at http://data.bls.gov/cgi-bin/surveymost?cu
Wage Index	Value of wages		BLS, Employment Cost Trends. Table 5 at http://www.bls.gov/web/eci/ecicois.txt